

SkillPicker: Tweezers for Recording and Training Dexterous Operations

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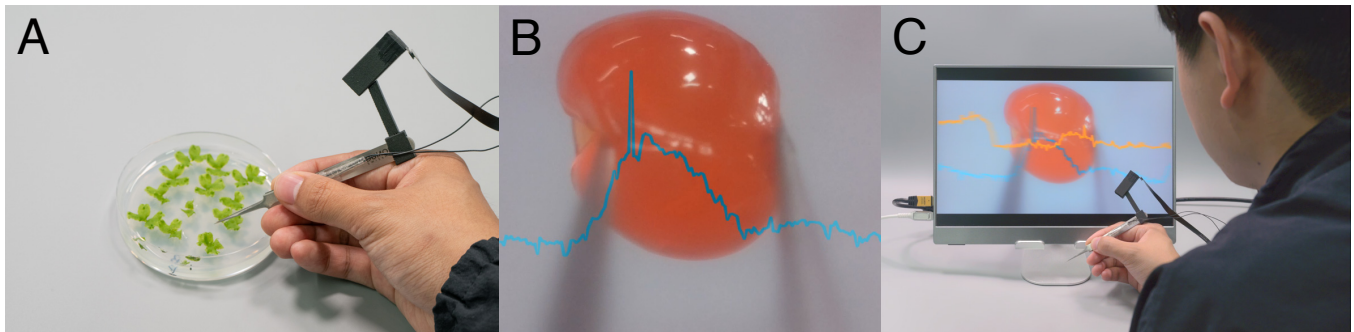


Figure 1: (A) SkillPicker assists unskilled trainees in improving force control accuracy while using tweezers. (B) Salmon roe captured by the magnifying camera. The blue line indicates the pressure applied by the thumb. (C) Beginner operator learning the appropriate amount of pinch force by comparing the real-time pinch force (orange) with the reference (blue).

ABSTRACT

Tweezers are used to carry out crucial functions in various research fields, most notably biology. Thus, proficiency in using tweezers impacts experimental outcomes. However, becoming proficient in using tweezers under expert guidance is not always easy for beginners because the force applied to the tip of a tweezer is virtually imperceptible. To help learners, we propose SkillPicker, which provides the user with images taken at the tip of their tweezers and measures the pinch force simultaneously. Thus, experts can record

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their technique using SkillPicker, enabling beginners to practice autonomously. SkillPicker is equipped with a small magnifying camera at the fulcrum of the tweezers and a pressure sensor to record the pinch force applied by the user's thumb. We establish an empirical correlation between the pressure applied and the pinch force at the tip, thereby enabling the estimation of the pinch force applied at the tip of the tweezers. SkillPicker also enables pressure values to be visualized or converted into musical notes on a scale to promote instant comprehension. This system facilitates the recording and sharing of tweezer operation, enabling beginners to learn dexterity both visually and physically.

CCS CONCEPTS

• **Applied computing** → *Interactive learning environments*; • **Hardware** → *Emerging interfaces*; • **Human-centered computing** → *Displays and imagers*.

KEYWORDS

Sensory sharing, Tweezers, Biotechnology, Teaching techniques

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1 INTRODUCTION

Tweezers are essential in medical and biological research—they are used for precise tasks that are difficult to accomplish with fingers. Operating a tweezer requires great expertise; e.g., experienced users often need to adjust their pinch force by sensing subtle changes in the specimen shapes. Therefore, continuous visual and tactile feedback is essential for the precise control of tweezers. Naturally, beginners often struggle to determine the optimal magnitude of force to be applied and execute dexterous techniques. To address these challenges, we propose an instructional tweezer-operation system called SkillPicker, which provides users with real-time video feedback of the tip, the magnitude of finger pressure, and recording functionality for this information. The accurate visual and tactile feedback provided by SkillPicker helps beginners master the skills required to operate tweezers competently, as depicted in Figure 1.

2 RELATED WORK

Existing literature has proposed quantifying dexterity while using tweezers, especially in the context of medical training. For instance, [Harada et al. 2015] proposed pinch force measurement using a strain gauge attached to the tips of tweezers to assess the skills of surgeons. Additionally, [Shalev 2020] proposed attaching a camera to tweezers to enable their use in concealed places, e.g., within the aural canal. We combine these approaches and propose a single tool that quantifies the user's pinch force while also allowing them to observe and record video images at the tip of the tweezers.

3 SYSTEM OVERVIEW

SkillPicker integrates a magnifying camera and a force-sensitive register (FSR) used as a pressure sensor. The camera captures a magnified view of the tip of a tweezer, which is displayed on a monitor. The pressure applied to the tweezer by the user's thumb is quantified by an FSR attached to the tool. These pressure values are plotted over the image obtained from the camera or converted into auditory feedback, e.g., musical notes on a scale. The presentation of the measured pinch force with real-time video assists users in acquiring the fine motor skills required to operate tweezers capably. The use of an eye-tracker with SkillPicker to track and measure gaze movement was also tested—we intend to explore this topic further in future research.

4 USER STUDY

We conducted a user study to investigate the efficacy of SkillPicker. Beginner tweezer operators were recruited and asked to perform a cut-and-place task on liverwort plantlets. This task is considered routine among plant biologists who work with liverwort. We

recorded an instructional video of the task being performed by a skilled plant researcher. Participants were randomly assigned to one of two groups. The control group watched the instructional video, while the other group (henceforth referred to as the SkillPicker group) watched an identical video that also included information on applied pinch force. Subsequently, the SkillPicker group was trained to apply or maintain the designated pressure while receiving visual and auditory pressure feedback, while the control group was trained to use conventional tweezers. Finally, both groups were asked to use SkillPicker without visual and auditory pressure feedback to complete the task.

During the training session, all participants who tested SkillPicker successfully applied and maintained the required force, suggesting that visual and auditory pressure feedback facilitates efficient skill learning. Participants were asked to rate the difficulty of the task on a scale of 1 to 7, with most rating it below 3. The results depicted in Figure 2 suggest that SkillPicker provides an effective instructional medium for learning tweezers operation.

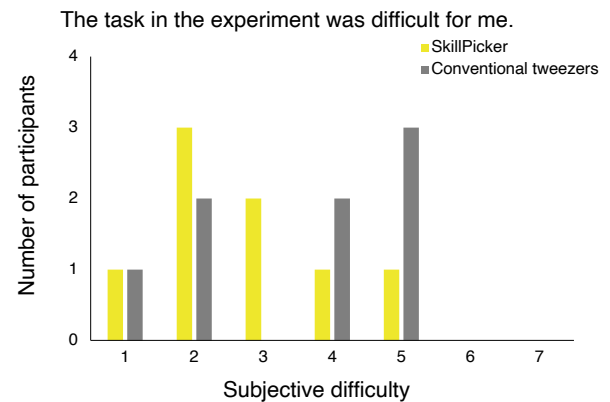


Figure 2: Comparison of the difficulty level of SkillPicker and conventional tweezers.

5 EXPERIENCE AT SIGGRAPH

We exhibit small specimens with varying degrees of hardness for the attendees to interact with using SkillPicker. Attendees are asked whether they are able to master control over the pinch force based on magnified real-time images and visual or auditory information representing the pressure being applied by their thumb.

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